

Abstract

The ability to detect thousands of planets orbiting other stars has enabled the examination of which planets outside our solar system most closely mirror Earth. For this study, data from NASA's Exoplanet Archive will be used to compare those rocky planets found around other stars using the Transiting Exoplanet Survey Satellite (TESS). Using Python, I developed a computational model that evaluates each planet according to four primary characteristics: planetary radius, density, equilibrium temperature, and stellar insolation. Each available measurement is compared to the corresponding Earth measurement and is assigned a similarity score, enabling a separate analysis of physical and environmental traits. All these attributes are then incorporated into a Composite Earth Similarity Score, with Physical- and Environmental-emphasis scores calculated to examine the sensitivity of the rankings. TOI-700 d is the top-ranked candidate, having a near-Earth radius, equilibrium temperature, and stellar insolation. These findings illustrate the need to consider multiple attributes rather than simply the size of the planet when determining whether an exoplanet can be considered Earth-like. However, this model measures similarity to Earth and not habitability, and therefore is limited in its ability to determine which planets may be capable of harboring life.

1. Introduction

The discovery of exoplanets changed the field of planetary systems research. Nowadays thousands of exoplanets have been discovered, including massive gas planets and small terrestrial exoplanets which are almost Earth-sized. The NASA Transiting Exoplanet Survey Satellite (TESS) greatly helped in the discovery of many planets since it observed nearby stars for the decrease in their luminosity due to passing planets (NASA, 2026). With the increase in the number of exoplanets discovered the main question of research changed from the presence of planets outside our solar system to the nature of those planets.

For example, one of the questions which scientists want to know is the existence of planets which are similar to Earth. Nevertheless, the determination of Earth-likeness is not as simple as it seems and does not only mean the size of the planet. An Earth-sized planet can be completely different in its composition and can have a completely different amount of energy from its star. Moreover, there is a difference between Earth-sized, Earth-like and potentially habitable planets, where the first one describes the size of the planet, second – multiple features and third – the ability to support life (Kopparapu et al., 2013; Rogers, 2015).

Thus, this study analyzes multiple characteristics of rocky planets discovered by TESS. Using data from the NASA Exoplanet Archive we compare planets based on four characteristics: planetary radius, density, equilibrium temperature and stellar insolation with their Earth counterparts. Instead of using any characteristic separately, we compare all of them with Earth

values to determine both physical and environmental similarities. The research question in this work is the following: which rocky planets discovered by TESS have maximum physical and environmental similarities with Earth? It is supposed that the rankings generated by multiple characteristics will differ greatly from rankings generated by planetary radius.

2. Scientific Background

Finding out how alike an exoplanet is to Earth requires taking into account a number of different physical and environmental features. This research focuses on four primary features of Earth-likeness: radius, density, temperature, and insolation. The orbital period is assessed as a secondary feature, but it is not taken into account when calculating similarity because of the heavy dependence of its importance on the characteristics of the planet's parent star.

2.1 Planetary Radius

Planetary radius provides a basic measurement of an exoplanet's size relative to Earth, which has a radius of one Earth radius. Radius is particularly useful for initially identifying planets that may be terrestrial rather than gaseous. Smaller planets are generally more likely to have rocky compositions, while sufficiently large planets are more likely to possess substantial gaseous envelopes (Rogers, 2015). For this reason, radius serves as the initial screening criterion in this study. However, similar size does not necessarily indicate similar composition. Two planets with nearly identical radii can have substantially different masses and internal compositions, meaning radius alone is insufficient to determine how closely an exoplanet resembles Earth (Chen & Kipping, 2017).

2.2 Planetary Density

Density provides further clues regarding the probable bulk composition of the exoplanet. The Earth has a density of about 5.51 g/cm^3 because of its largely rocky and metallic nature. As a measure of both mass and volume, density offers data which are not available from just looking at the radius. A density close to that of the Earth implies a mostly rocky makeup of the planet, while a lower density suggests a higher proportion of volatile and gaseous materials (Chen & Kipping, 2017). Thus, density plays a significant role in assessing physical resemblance by enabling the distinction between Earth-size planets and those which have Earth-like bulk compositions.

2.3 Equilibrium Temperature

Equilibrium temperature estimates a planet's temperature based on the balance between incoming stellar radiation and the energy the planet radiates back into space. Earth's equilibrium temperature is approximately 255 K. This value is different from Earth's average surface temperature, which is higher because the atmosphere traps heat through the greenhouse effect. An exoplanet's equilibrium temperature therefore does not represent its actual surface temperature, particularly when its atmospheric composition is unknown. Nevertheless, it provides a standardized method for comparing the approximate thermal environments of different planets. In this study, equilibrium temperature is used as an environmental similarity criterion while recognizing that atmospheric conditions could cause actual surface temperatures to differ substantially.

2.4 Stellar Insolation

Insolation refers to the amount of solar energy that an exoplanet receives from the star it orbits. The Earth represents 1 unit of insolation, providing a basis against which exoplanets from other stars can be compared. Insolation affects the temperature of the planet greatly and is particularly important in determining whether liquid water can exist on a planet under the right atmospheric conditions (Kopparapu et al., 2013). A planet that receives much higher levels of energy from the star than Earth is going to be much hotter while one with lower energy is much cooler.

2.5 Orbital Period

Orbital period represents the time it takes a planet to complete a single circuit around its parent star. While the orbital period of Earth is roughly 365.25 days, being close to this number does not necessarily mean a presence of an Earth-like environment. The orbital period is determined by both the length of the semi-major axis of the orbit and by the mass of the parent star. Importantly, however, the distance at which the planet receives an Earth-like level of energy from the star is set by the star's luminosity (Kopparapu et al., 2013).

Since planets circling smaller and less luminous stars will be much closer to their parent stars in order to receive an amount of energy from the star equal to Earth, such planets will have a shorter orbital period than Earth. On the other hand, more luminous stars will allow the planet to be farther away from the star in order to receive a similar amount of energy and therefore have a longer orbital period. As a result, a planet that has a short orbital year may receive an amount of energy from the star that is similar to Earth's, while having an orbital period of 365 days might mean something else entirely.

3. Data

3.1 Data Source

The planetary data used for the purposes of this research were gathered from NASA's Exoplanet Archive, an online repository holding confirmed exoplanets and the physical and orbital parameters related to those planets. Planetary data was gathered from the "Planetary Systems" table on the NASA website and filtered to only include confirmed planets discovered by the Transiting Exoplanet Survey Satellite (TESS).

3.2 Sample Selection

Since there are planets in a variety of sizes including small rocky planets and huge gas giants, the data set was confined only to those planets which could be considered rocky. Planetary radius became the first basis for selection, where the planets that had radius equal to or less than 1.8 Earth radii were selected as part of the data set. This was done to avoid selecting the huge planets which might have big gaseous envelopes while keeping the smaller ones which might have rocky composition. However, planetary radius was used only as an initial basis of selection rather than proof of being rocky.

3.3 Variables

For each planet, the data set includes the values of the planetary radius, mass, density, orbital period, semi-major axis, equilibrium temperature, stellar insolation, and distance from the Earth. Planetary radius and mass are given in relation to the corresponding parameters of Earth, while density is given in g/cm³. Orbital period is given in days, semi-major axis – in astronomical units, and equilibrium temperature – in Kelvin degrees. Insolation is given in relation to the stellar radiation on Earth. Of these variables, radius, density, equilibrium temperature, and insolation are classified as four main variables of the Composite Earth Similarity Score, while the others are kept as secondary variables.

3.4 Missing Data

Not all exoplanets in the NASA Exoplanet Archive had measurements for all the variables. Thus, when there was no measurement, the variable was marked as not available and not as a zero. The importance of this aspect is that a zero would mean that there was a measurement, while in fact, there was not any. In similarity charts, if the planet did not have the

needed measurement for the calculations, it was written as "Not calculable." In composite analysis, data availability was considered independently to prevent some planets with few measurements from being too similar to Earth.

4. Computational Methodology

4.1 Data Processing

The exoplanet data set was processed using Python. The pandas package helped with importing the NASA Exoplanet Archive data from an Excel file, finding any missing values, and organizing the planetary measurements for further analysis. The matplotlib.pyplot package was used for creating graphs that showed the comparison of each planet with Earth. This included plotting the raw planetary measurements, the Earth similarity percentage per planet, and finally, the Composite Earth Similarity Score. The order of planets was preserved throughout all graphs to make the comparison easier.

4.2 Earth Reference Values

Each exoplanet was compared to Earth using standardized parameters for Earth. Earth was taken as a sphere having radius equivalent to 1 Earth radius, having density of 5.51 g/cm³, temperature around 255 K and solar flux equal to 1 Earth insolation. This was used to calculate the degree of resemblance of the exoplanets on each of the four primary criteria.

4.3 Individual Similarity Metric

For each criterion, the individual measurements were assessed by comparing them to the values on Earth and calculating how different they were in relation to Earth. The difference was then converted into the similarity measure. If the measurement of a planet had been equal to that of Earth, the planet would have received 100% in similarity. The higher the difference from Earth, the lower the similarity value got, down to 0%.

In the computation used, the absolute difference between values of the planets and the reference value for Earth was considered; therefore, deviations both above and below the reference value were given equal weighting. For example, a planet that is 20% different from the reference value of Earth, regardless of direction, would receive an equal score for that criterion. This makes it possible to have a consistent scale for comparisons of measurements in different units and different number ranges on a single model.

4.4 Individual Similarity Indices

For each of the planets considered, there have been calculated four different similarity indices: radius similarity index, density similarity index, equilibrium temperature similarity index, and insolation similarity index. Radius and density comprise the physical part of the evaluation, while equilibrium temperature and insolation are the environmental ones. This separation allows one to understand why a planet is ranked as it is and to identify the cause behind its high or low rating.

4.5 Composite Earth Similarity Score

The four measures of similarity have been combined together into one Composite Earth Similarity Score. In the primary model, all four measures had the same weightage of 25%. This means that all physical and environmental factors carried equal importance to the overall score. Therefore, for a planet to have a high rank, it has to be similar to Earth in many factors, not just one measure.

Orbital period was left out of the composite formula due to the dependence of its environmental importance on the nature of the parent star. Insolation and equilibrium temperature provide better predictors of the energy conditions of the planet. Planet mass was not used as another factor since it influences the planet's density along with other aspects of its nature.

Two additional models have been constructed in order to determine the effect of the weighting system used. In the first, the environmental emphasis model, 35% has been assigned to equilibrium temperature and another 35% has been assigned to insolation, while 15% has been allotted to radius and 15% to density. In the second, the physical emphasis model, 35% has been assigned to radius and 35% to density, while 15% has been allotted to equilibrium temperature and insolation. Comparing the rankings produced by these three models provided a sensitivity analysis of how dependent the results were on the weighting assumptions.

4.6 Data Completeness

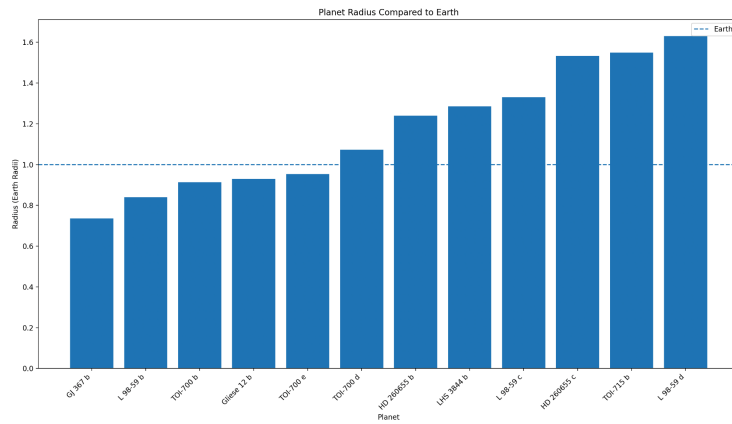
Similarity score calculation was possible only if there was a value in the NASA Exoplanet Archive for that measurement. Missing values were not considered to have a similarity score of 0%, since this assumption would falsely represent missing data as being different. The "Not calculable" category was used instead.

Completeness of data was especially important in the case of the composite analysis since an average of only the data that is available can cause an overestimation of the rank of a planet that has incomplete data. In order to distinguish planets that do not have all of the measurements from those that have them, a composite score that can be directly compared was determined.

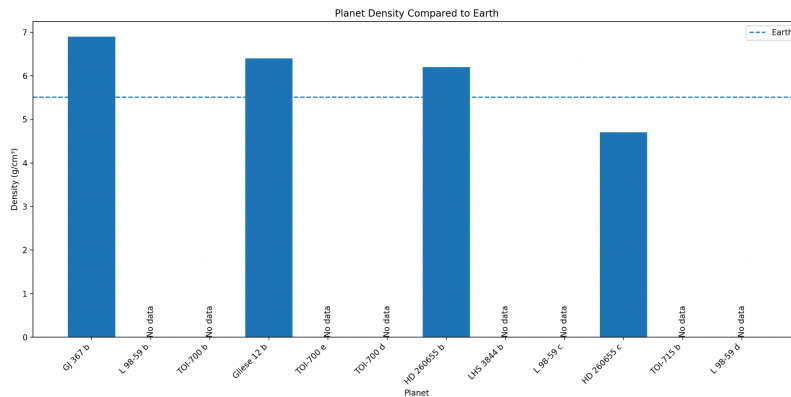
5. Results

5.1 Raw Physical Comparisons

The planetary radii showed rather low variations relative to the rest of physical parameters in the given sample. All twelve planets had their radii smaller than about 1.65 Earth radii, which can be explained by the fact that radius was used as a parameter in creating this set of planets. Particularly, TOI-700 e, TOI-700 d, Gliese 12 b, and TOI-700 b had the smallest difference from Earth's radius among the other planets. TOI-700 e had ≈ 0.95 Earth radii, Gliese 12 b had ≈ 0.93 Earth radii, while TOI-700 d had ≈ 1.07 Earth radii.

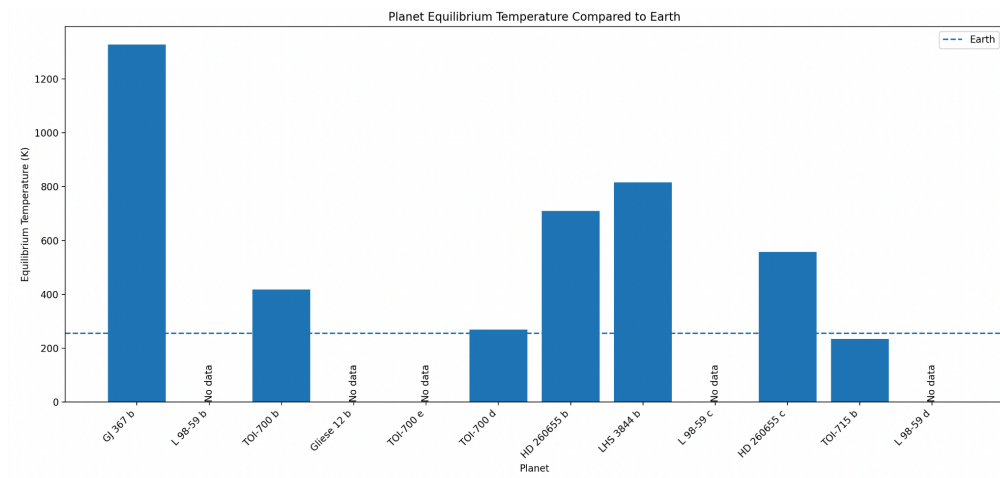


Measurements of density were significantly less complete compared to measurements of radius. Density values are only known for four planets: GJ 367 b, Gliese 12 b, HD 260655 b, and HD 260655 c. It should be noted that all four have densities close to that of Earth, which is equal to 5.51 g/cm^3 . The density of HD 260655 b is around 6.2 g/cm^3 , whereas the density of HD 260655 c is slightly lower and equals 4.7 g/cm^3 .

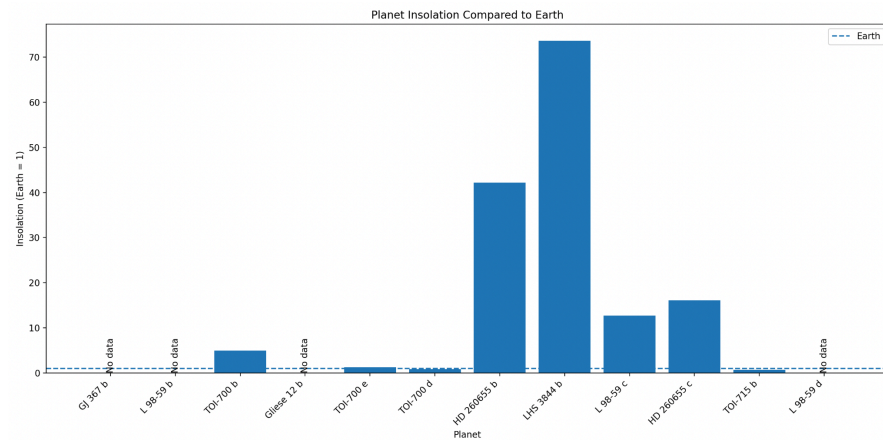


5.2 Raw Environmental Comparisons

Equilibrium temperature proved more variable than the planetary radius. The planet GJ 367 b had the highest equilibrium temperature, surpassing 1,300 K, and several other planets, such as HD 260655 b and LHS 3844 b, demonstrated a higher equilibrium temperature than Earth's, which is around 255 K. On the contrary, TOI-700 d and TOI-715 b appeared quite similar to Earth from the standpoint of temperature; their equilibrium temperatures reached approximately 269 K and 234 K correspondingly. Thus, there is no connection between the similarity in size and temperature of the planets.

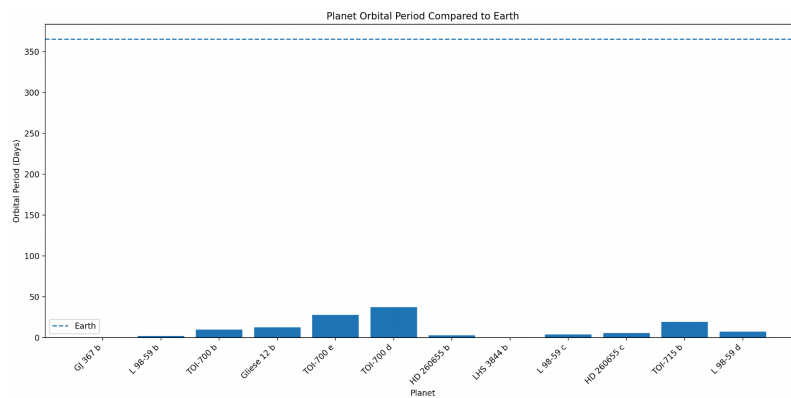


A similar trend is observed in the case of stellar insolation. Several planets receive considerably more energy from their parent stars than the Earth. Particularly, LHS 3844 b and HD 260655 b receive about 70 and 40 times the amount of insolation received by Earth, respectively. However, TOI-700 d, TOI-700 e, and TOI-715 b have an insolation that is much closer to that of Earth. Amongst them, TOI-700 d receives about 0.85 times the insolation of Earth, making it the nearest Earth equivalent.

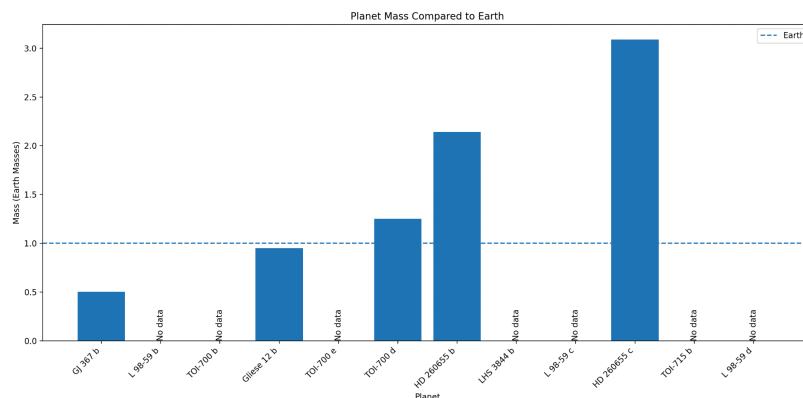


5.3 Supplemental Comparisons

The orbital periods displayed considerable variations when compared to Earth's 365.25-day period. In this study, each planet took less than 40 days to complete an orbit, and there were some cases of just a couple of days or less to complete an orbit. The planet TOI-700 d had the longest orbital period of all the planets in this study, which was 37.4 days. It is interesting to note that despite having an orbital period just one tenth that of Earth, TOI-700 d was among the most Earth-like planets with respect to the equilibrium temperature and insolation. This shows a clear picture of why orbital period is not an effective independent metric of environmental similarity. Planets orbiting weaker stars can receive near identical amounts of stellar energy while maintaining a much closer orbit.

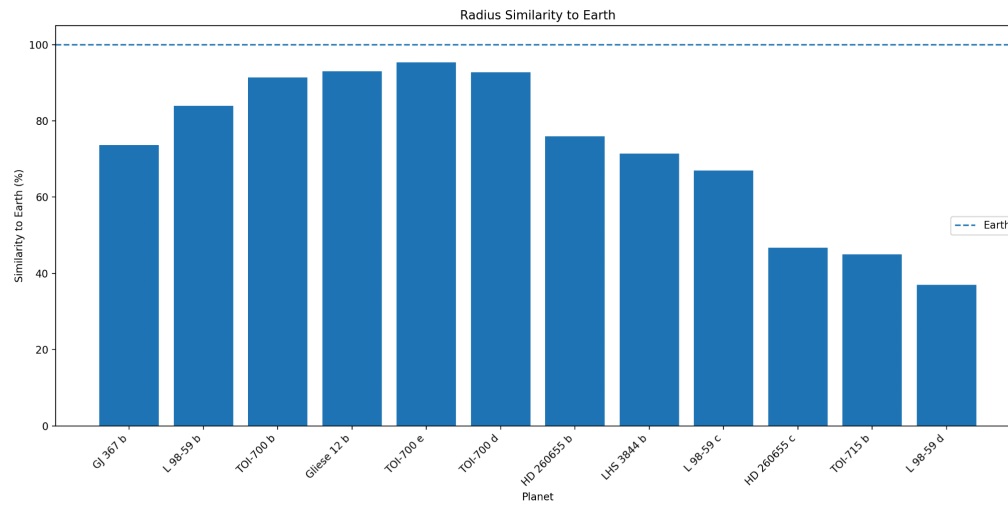


Data on mass was not available for many planets. For those planets where mass data was available, the masses varied between 0.5 times the mass of Earth, in the case of GJ 367 b, to more than three times the mass of Earth in the case of HD 260655 c. Gliese 12 b had a mass similar to that of Earth, which was 0.95 times the mass of Earth, whereas TOI-700 d had a mass of 1.25 times the mass of Earth. Mass was used as supplemental information rather than incorporated into the Composite Earth Similarity Score because of its relationship with planetary radius and density.

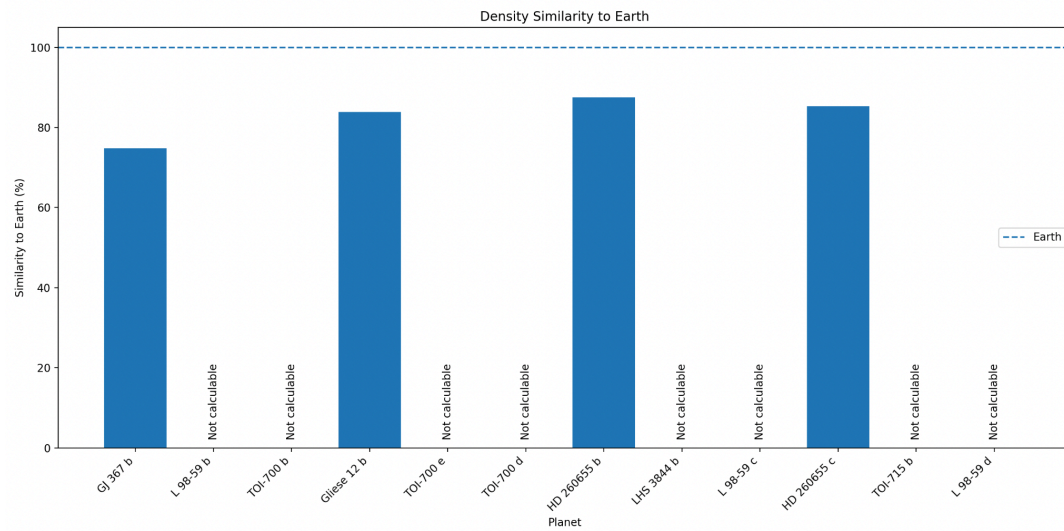


5.4 Individual Similarity Scores

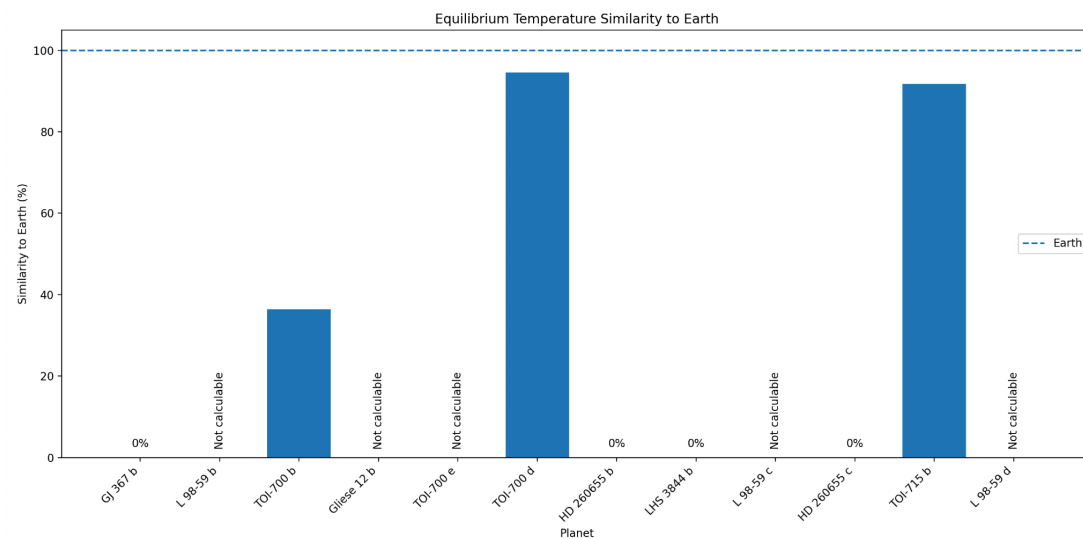
Many of the planets had good radius similarities as well. Notable were TOI-700 e, which showed the most similarity in terms of radius at about 95% similarity, as well as Gliese 12 b and TOI-700 d, whose radius similarities were over 90%. Even the planets with radii less similar to that of the Earth still had decent similarities.



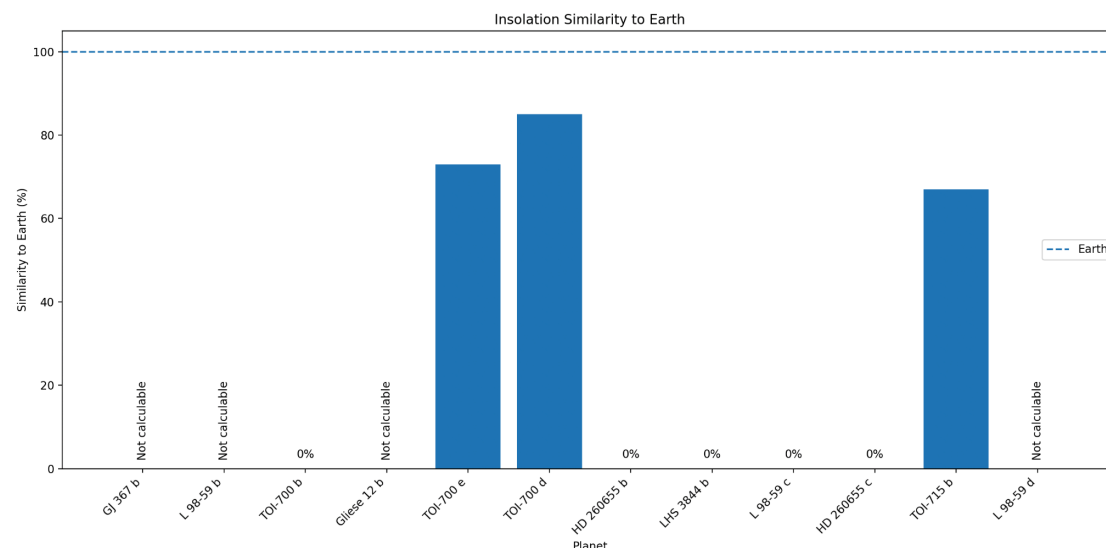
The similarity in densities could only be found among four planets. HD 260655 b had the most similar density of about 88%, whereas HD 260655 c and Gliese 12 b had similar densities of about 85% and 84%, respectively. GJ 367 b was the least similar planet in terms of density, having an estimated density similarity of about 75%. The remaining planets were classified as “Not calculable” because no density measurement was available.



There was more dispersion in temperature of the planets for the equilibrium temperature calculation. TOI-700 d had a similar temperature of almost 95%, while TOI-715 b had 92% similarity in temperature. On the other hand, TOI-700 b had a much lower similarity percentage of 36%. There were some planets that reached the 0% lower limit of similarity calculations. Some other planets were considered “Not calculable,” since there was no data on equilibrium temperature for such planets.

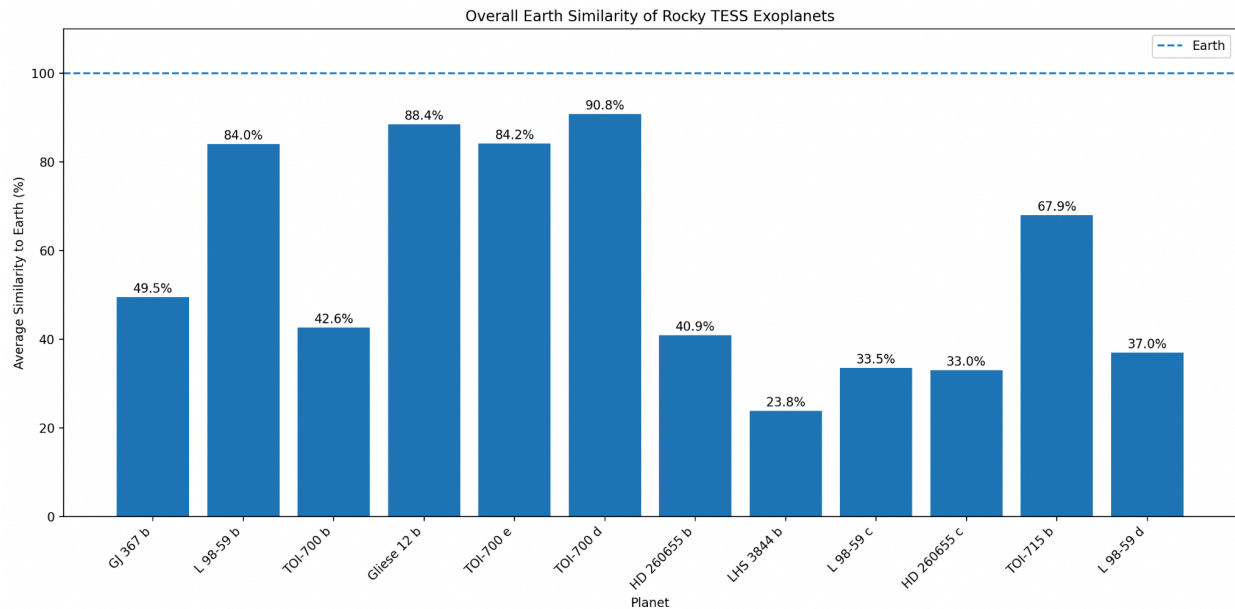


TOI-700 d had the most similar insolation, at around 85%. Next came TOI-700 e, which showed similarity at around 73%, followed by TOI-715 b, at around 67%. Planets with stellar radiation that is much higher than that experienced on Earth had the lowest similarity of 0%. Similar to density and equilibrium temperature, some planets could not be analyzed since there was no data available for certain parameters.



5.5 Composite Results

With regard to the measurements that were available, TOI-700 d scored the highest initial Composite Earth Similarity Score with a score of 90.8%. Coming second was Gliese 12 b with an index score of 88.4%. In third place was TOI-700 e followed by L 98-59 b with scores of 84.2% and 84.0%, respectively. TOI-715 b had a lower index score of 67.9%, despite performing strongly in equilibrium temperature and insolation, largely because its radius was substantially greater than Earth's.



Given the cumulative findings, TOI-700 d emerged as the most promising candidate given the physical and environmental data provided. The planet had a radius measuring about 1.07 times the radius of the Earth, thus having a similarity in radius of about 93%. Additionally, it had an equilibrium temperature of about 269 K, which translated into a similarity in equilibrium of about 95%, and it had about 0.85 of the Earth's stellar irradiance, which gave a similarity of about 85%. Unlike other planets in the dataset, TOI-700 d thus showed similarity in physical and environmental aspects rather than size only.

However, the interpretation of the composite ranking should take into account the issue of data availability. Four main indicators were not equally present for all planets under consideration, and there was no density indicator for TOI-700 d. Thus, the 90.8% score cannot be interpreted as a four-variable evaluation of the Earth similarity rating. Additionally, there are planets like L 98-59 b that received relatively high initial ratings while having much less available data than TOI-700 d. Thus, TOI-700 d can only be called the best candidate given the available information, and not the planet that is 90.8% Earth-like. This distinction also reinforces

that the model measures Earth similarity within the available dataset, not planetary habitability or the probability of supporting life.

6. Sensitivity Analysis

6.1 Purpose

A sensitivity analysis was conducted to determine how much influence the assumptions about the weights built into the model have on the final rankings. At first, the Composite Earth Similarity Index assumed that radius, density, equilibrium temperature, and insolation from the star had the same weight. However, the model cannot provide any objective reasons for assigning higher priority to physical similarities than to environmental ones. Thus, two additional methods of weighting were considered to determine whether emphasizing different types of characteristics would substantially change the results.

Since some planets did not have one of the four key measures, at least three measures were needed in order to assign a score to a planet. In case a measure was missing, the weightages of the other measures were normalized in proportion. Planets that had less than three measures were classified as “Not calculable.”

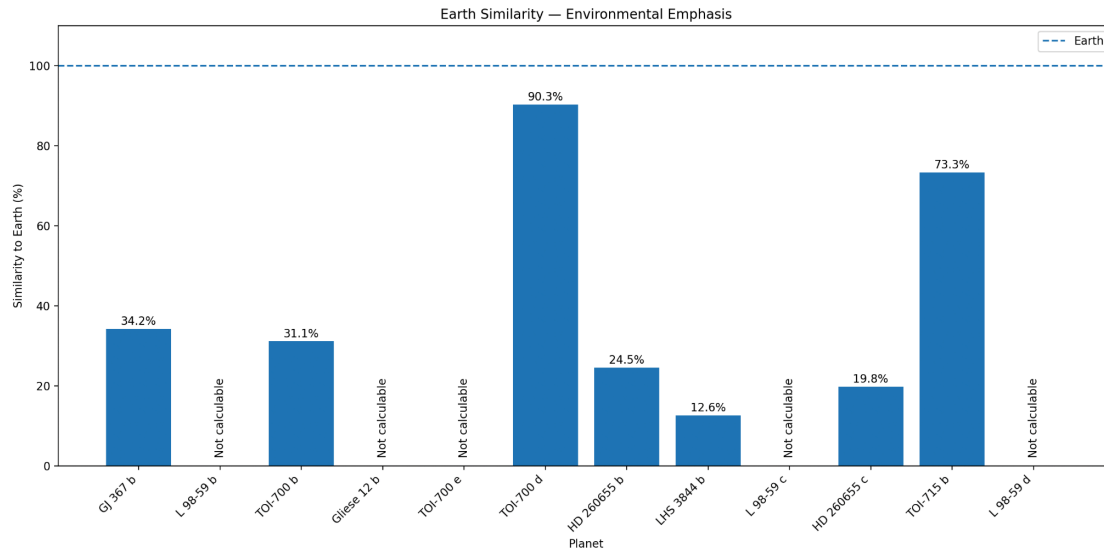
6.2 Model 1: Equal Weighting

The first weighting gave all four factors equal consideration by weighting each one by 25%. Physical parameters such as radius and density thus had a cumulative weight of 50%, while environmental parameters like equilibrium temperature and insolation accounted for another 50%. Using these weights, TOI-700 d scored highest at around 90.8%, making it the strongest candidate based on the available physical and environmental evidence.

6.3 Model 2: Environmental Emphasis

The environmental-emphasis model increased the weights of equilibrium temperature and insolation to 35% each while reducing radius and density to 15% each. This gave environmental characteristics 70% of the total weight and physical characteristics 30%.

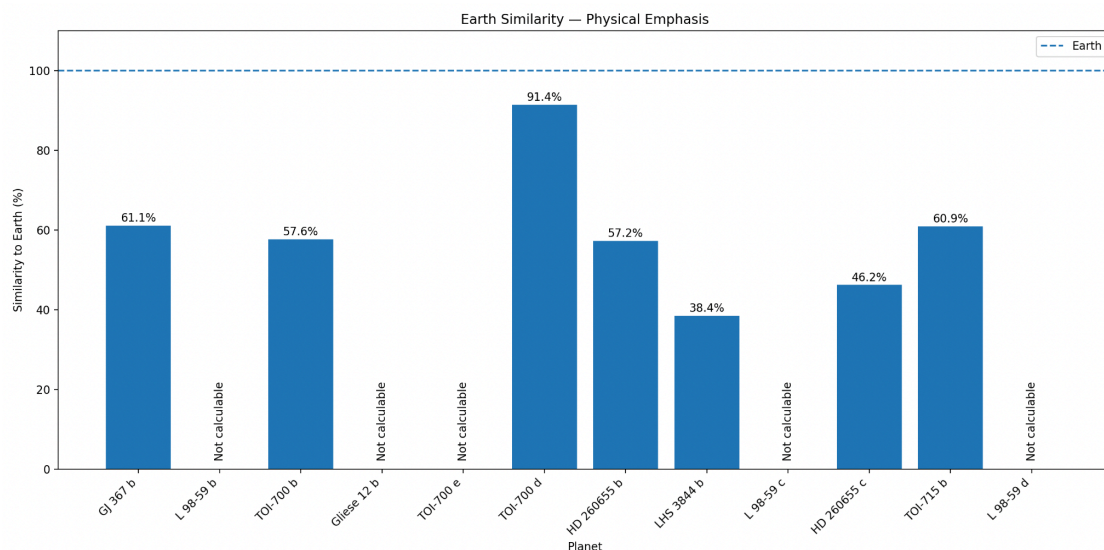
The planet TOI-700 d scored the most out of all the planets with a rating of around 90.3% of environmental emphasis. After that came TOI-715 b which scored around 73.3%. However, planets which experience extreme heat or stellar radiation have seen a considerable fall in their scores. For example, HD 260655 b scored just 24.5% even though it has strong physical parameters, because its equilibrium temperature and insolation were substantially different from Earth's.



6.4 Model 3: Physical Emphasis

The physical-emphasis model reversed the weighting system, assigning 35% each to radius and density and 15% each to equilibrium temperature and insolation. Physical characteristics therefore represented 70% of the weighting, while environmental characteristics represented 30%.

TOI-700 d once again came out on top, increasing marginally to around 91.4%. The other two planets that scored relatively high were GJ 367 b and TOI-715 b, which scored 61.1% and 60.9%, respectively. However, there were some planets that scored low using the environmental criteria but improved significantly when physical criteria were emphasized more. This includes GJ 367 b, whose score increased from 34.2% to 61.1%.



6.5 Model Comparison

The modification in the weighting system has caused some significant differences in the scores of certain planets. The Earth-similarity ratings are partly dependent on the way physical and environmental parameters are weighted. The planets that have physical similarities with earth but extreme environment received high ratings due to the physical system model, while the planets that had Earth-like temperature and solar radiation received high ratings due to the environmental model.

TOI-700 d was found to be the notable outlier in the scoring system. TOI-700 d was at the number one position in all the three systems, having 90.8%, 90.3%, and 91.4% in the equally-weighted, environmentally-weighted, and physically-weighted models respectively. The difference in its rating in all three weighting schemes is just 1.1%. This stability indicates that its high ranking was not dependent on one particular weighting assumption.

In light of the sensitivity analysis, TOI-700 d is clearly the best choice of all exoplanets based on the physical and environmental data that is currently available. Although the exact numerical ratings may depend on the model used and the data collected, TOI-700 d remained number one even when there were significant changes in the weights given to physical and environmental parameters. Other planets showed more dependency on the model used, indicating that their apparent Earth similarity was more strongly associated with particular categories of measurements.

7. Discussion

7.1 Most Earth-Like Candidates

From among the results obtained, TOI-700 d comes out as the strongest contender based on the available information regarding both physical and environmental parameters. In the equal weight model, TOI-700 d had the highest Composite Earth Similarity Score of about 90.8%. The major contributing factors were its equilibrium temperature and radius similarity score which were estimated to be about 95% and 93%, respectively. Insolation was also fairly similar to that of Earth being around 85%. The planet did not have a density score but was still able to achieve high similarity in all the three criteria in which it had information.

There are other planets which had some very similar qualities to Earth in one of the dimensions but had to be left out due to lack of information or weak performance in other criteria. Gliese 12 b had a radius and measured density values relatively closer to that of Earth but since there was no data concerning the equilibrium temperature and insolation of this planet, it was impossible to make an environmental comparison. TOI-700 e had the highest radius similarity among the planets and its insolation was quite Earth like but due to lack of density and

equilibrium-temperature, it could not meet the three measurements criterion. TOI-715 b, meanwhile, had strong environmental characteristics, including an equilibrium temperature of approximately 234 K and insolation of approximately 0.67 Earth units, but its larger radius reduced its physical similarity.

These variations show that the result should not be seen as a direct comparison of the size of planets to that of Earth. The best choice turned out to be TOI-700 d since all the evidence pointed towards similarity in its physical and environmental features.

7.2 Importance of Multivariable Analysis

The comparison between single variables and compound scores illustrates the necessity to take into account several properties of a planet to evaluate its Earth similarity. The radius itself generated several high-similarity candidates: for instance, TOI-700 e with radius of 0.95 Earth radii could be considered more similar to Earth in terms of radius than TOI-700 d. Nonetheless planetary radius is not an indicator of environmental characteristics.

It became especially obvious for planets which were rather Earth-like in size, yet had extreme environments. For instance, GJ 367 b had a radius of 0.74 Earth radii and an equilibrium temperature above 1,300 K. In the same way, HD 260655 b had rather Earth-like physical properties (including the density) but received more than 40 times of Earth's insolation. LHS 3844 b had a relatively small radius and received more than 70 times of Earth's insolation. Thus, the classification of a planet as small/rocky does not guarantee an Earth-like environment.

In some cases, it was the other way round. TOI-715 b was much larger than Earth, having approximately 1.55 Earth radii, thus having low Earth similarity of radius; however, its equilibrium temperature and insolation were much closer to Earth's values than those of many smaller planets. The use of multiple variables, thus, reinterpreted the dataset, separating physical similarities from environmental ones. The initial hypothesis that a multi-dimensional approach provides different results from a single-variable comparison was confirmed.

7.3 Earth Similarity V. Habitability

The Composite Earth Similarity Score cannot be interpreted as a measure of habitability. The model quantifies the degree to which the selected parameters of planets approximate terrestrial values. For instance, a score of 90% means similarity according to the existing characteristics and the applied model in this study; this score does not mean the 90% chance of a planet being habitable.

There are numerous other characteristics that influence habitability but they are not included in the model. Atmospheric composition and pressure significantly influence surface temperature due to the greenhouse effect, which means that having an Earth-like equilibrium temperature does not necessarily imply having the same temperature on the planet's surface. Also, such characteristics as having liquid water, stellar activity, eccentricity, magnetic fields,

geological processes, as well as the stability of climate can significantly influence whether the planet can have habitable conditions.

Therefore, the identification of TOI-700 d as the best candidate based on the available physical and environmental data should be considered within the scope of this research. TOI-700 d turns out to be the closest approximation to the Earth according to the available characteristics in the model. The identification of TOI-700 d as the closest approximation to the Earth in no way confirms its habitability, possession of liquid water or existence of life there.

8. Limitations

8.1 Data Limitations

The main limitation of this study is associated with the absence of some data on the analyzed exoplanets. While the planetary radius and orbital period were accessible for almost all planets in the sample, other measurements such as the planet's mass, density, equilibrium temperature, and insolation were not accessible for some planets. In particular, obtaining the density was especially challenging since it was necessary to know the mass and radius of the planet. Hence, there were some planets that could not be evaluated according to all four criteria. Nevertheless, the application of the requirement of at least three measurements and weight normalization reduced the effect of the incomplete data.

The data themselves are associated with the observational errors and different levels of precision. Usually, exoplanet properties are estimated based on the observations; therefore, estimations might change with time based on new data. Besides, the atmospheric data of the planets were very limited, which makes it hard to compare the surface conditions of the exoplanets.

Besides the observational limitations described above, the sample is affected by the observational bias associated with the TESS mission and the transit method itself. In general, the planets that transit their star more often are easier to find, which means that there is an intrinsic bias towards shorter orbits. Consequently, the sample of planets cannot be considered representative.

8.2 Model Limitations

The Earth Similarity Index developed through this project is a customized comparison index rather than a scientifically validated index of planetary habitability. The index formula assumes that the similarity value decreases in a linear fashion as the measurement departs from the corresponding Earth's value, with a minimum limit of 0%. While this provides a consistent

system for comparison of a variety of variables, actual planetary systems do not have to react linearly to the variations in radius, density, temperature, and insolation levels.

The main composite index model gives equal weight to all four categories. Despite the sensitivity analysis performed on various weighting scenarios favoring physical or environmental aspects, these ratios remain to be assumptions and are not proven scientifically as indicators of the actual weight of each particular characteristic. Besides, the equilibrium temperature and stellar insolation are physically interconnected, which means that taking into account both of them might result in the overrepresentation of a planet's thermal environment according to the assigned weighting percentages.

Finally, condensing a variety of physical and environmental features into numerical values loses some of the information content. Two planets with identical composite similarity indices can be similar because of completely different values of physical and environmental characteristics. Also, this model has an inherent Earth-centric approach, which assumes that higher similarity to Earth is scientifically relevant, although other non-Earth-like environments might be important for habitability studies.

8.3 Unmeasured Habitability Factors

Another limitation of this study is that many factors related to the actual habitability of a planet could not be included in the model. In particular, very little is known about the atmospheres of most of the exoplanets included in the sample. Atmospheric composition and pressure can significantly affect the temperature of a planet through processes such as the greenhouse effect. Therefore, even if a planet has an equilibrium temperature similar to Earth, its actual surface temperature could be substantially different. Similarly, receiving an amount of stellar insolation similar to Earth does not necessarily mean that a planet has liquid water or Earth-like surface conditions.

There are also several other characteristics that may influence habitability but were not considered in the Earth Similarity Index. These include orbital eccentricity, stellar activity and flares, magnetic fields, geological activity, tidal locking, and long-term climate stability. For example, a planet orbiting an active star could experience intense stellar radiation even if its equilibrium temperature and insolation appear favorable. Likewise, tidal locking could produce major temperature differences between the side of the planet facing its star and the side facing away from it.

Consequently, the Earth Similarity Index should not be interpreted as a measurement of whether a planet is habitable or capable of supporting life. Instead, it measures how closely each planet resembles Earth according to the limited physical and environmental characteristics available in the NASA Exoplanet Archive. Therefore, TOI-700 d can be identified as the strongest candidate based on available physical and environmental evidence, but this does not confirm that it has Earth-like surface conditions or that it is actually habitable.

9. Conclusion

This research aimed to find out which of the discovered TESS rocky planets are most similar to Earth. The analysis of their physical and environmental characteristics showed that no single characteristic defines the planet's similarity to Earth. While some planets were the closest to Earth in terms of density, radius, or equilibrium temperature, their rankings shifted when considering all characteristics in the composite score.

The analysis of the Composite Earth Similarity Value (ESV) indicated that TOI-700 d was the strongest candidate based on the available physical and environmental evidence. TOI-700d is an Earth-sized exoplanet orbiting within the habitable zone of the red dwarf star TOI-700, located about 100 light-years away in the constellation Dorado. Its radius is relatively close to Earth's, while its equilibrium temperature and stellar insolation are also among the most Earth-like in the sample. The sensitivity analysis conducted corroborates this result, with TOI-700 d being the highest ranking in all the scenarios. Thus, the choice of TOI-700 d was not purely dependent on the choice of weights in the composite ESV model.

The orbital period was excluded from the composite ESV due to its dependence on the host star. A planet orbiting a smaller, cooler star can receive Earth-like levels of stellar radiation while orbiting much closer to its host star, resulting in a substantially shorter orbital period. Therefore, a direct comparison of orbital period with Earth's is not meaningful without considering the properties of the host star. For this reason, equilibrium temperature and stellar insolation were used for the comparison.

However, it should not be assumed that these similarities indicate the possibility of life on these planets. The model is limited by the current available data and does not account for various crucial factors. In addition, the model is based on Earth's characteristics as the sole reference point. Therefore, the composite ESV is a useful tool for finding the planets most similar to our own, but its application should be limited to these purposes. As more planets are found and characterized, such a model can be beneficial in identifying new planets analogous to Earth.

10. References

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